



RoboVitals AI — AI-Powered Health Insights

Instructions for Use

Camera-based pulse and respiratory rate measurement

HELLOGARD INTELLIGENCE · CONFIDENTIAL — FOR PATIENT AND CLINICAL REVIEW

| Glossary of Symbols

SYMBOL	SYMBOL TITLE	SYMBOL DESCRIPTION
	Consult Instructions for Use	Indicates the need for the user to consult the Instructions for Use.
	Caution	Indicates that caution is necessary when operating the device or control close to where the symbol is placed, or that the current situation needs operator awareness or operator action in order to avoid undesirable consequences.
	Manufacturer	Indicates the medical device manufacturer.
	Prescription Use Only	Caution: Federal (US) law restricts this device to sale by or on the order of a physician. Indicates the product is indicated for professional use only.

| 1. Getting Started

What is RoboVitals AI?

RoboVitals AI is a web application that measures your vital signs using the camera on your smartphone, tablet, laptop, or desktop computer. RoboVitals AI can be configured as a General Wellness application and/or prescribed by your healthcare provider as a Software as a Medical Device (SaMD).

ICON	PRODUCT USE	DESCRIPTION
	Prescribed by Healthcare Provider (Rx Only)	You can identify the prescription measurement display by the  icon on pulse rate and respiratory rate results. Tap Submit to send results to your healthcare provider.
	General Wellness (non-prescription)	You can identify general-wellness vitals (such as blood pressure or oxygen) by the  icon on those result cards. These readings are for information only and do not replace medical advice.

RoboVitals AI is a browser-based application provided through your healthcare organization. Open the application link in a supported browser on your smartphone, tablet, or computer. The link does not expire after 72 hours — you can return to the same application URL whenever you need to measure your vitals, unless your provider gives you a different access instruction.

How it Works

Each time your heart beats, your face shows tiny color changes that are invisible to the human eye. Similarly, each time you breathe, your shoulders and torso move slightly. RoboVitals AI uses your camera to detect subtle changes in facial coloration and shoulder motion to calculate pulse rate (PR) and respiratory rate (RR) in real time.

Before You Measure — Patient Identification

1. Open the RoboVitals AI application link in your browser.
2. Wait for **AI-Powered Vital Monitoring** to finish loading if a progress bar is shown.
3. Enter your **Patient ID** provided by your healthcare organization, or tap **Generate ID** if instructed to create a new ID.
4. The app automatically looks up your patient record.
 - **Existing patient:** When your record is found, tap **Capture Vitals**.
 - **New patient:** Complete the registration fields (email, name, date of birth, height, weight, and temperature), then tap **Capture Vitals** to create your record and continue.
5. Tap the **Help** button at any time to open these Instructions for Use.

Measurement Session Overview

1. On the **Health Insights** screen, allow camera access when your browser prompts you.
2. Center your face in the on-screen guide and tap **Start Scan**.
3. Hold still while the scan runs (about 60 seconds). A progress indicator shows scan status.
4. When the scan finishes, review your pulse and breathing results on screen.
5. If prompted between attempts, choose **Retry** to measure again or **Submit** to keep partial results you already have.
6. After all scan attempts, review your results and tap **Submit** to send them to your healthcare provider, or **Discard these results and restart scan** to start over.
7. After a successful submit, review the results summary and tap **Done** to return to the home screen with your Patient ID retained for another session.

NOTE: Tap **Back** at any time to exit and return to the home screen. This clears your current session. Use **Done** after a successful measurement to start another session for the same patient.

2. Instructions for Use

Read this manual carefully and pay attention to the warnings and cautions throughout.

Device Description

RoboVitals AI is a software application that utilizes existing optical camera technology embedded in a smartphone, tablet, or computer to estimate vital signs such as pulse and respiratory rate from a video recording.

Indications for Use

RoboVitals AI is intended for non-invasive spot measurement of pulse and respiratory rate of adult patients in home use, hospitals, clinics, and long-term care settings.

RoboVitals AI is indicated for use on adults 22 years of age or older who do not require critical care or continuous vital signs monitoring.

RoboVitals AI is not intended to direct therapy independently.

Contraindications

The following facial conditions may impact the ability to measure your vital signs. Check with your healthcare provider if you have any of the following:

- Scars or burns
- Port wine stains or large birthmarks
- Tattoos or henna
- Rosacea, eczema, and rashes
- Bruising

Warnings and Precautions

WARNING: RoboVitals AI should be used under the care of your physician. Always consult your physician regarding clinical decisions. Do not rely on RoboVitals AI as the only basis for medical decisions; pulse and respiratory rate measurements alone are not intended to direct medical care.

CAUTION: To ensure proper operation, avoid distractions. Do not use RoboVitals AI while driving, walking, or participating in any activity requiring your full attention.

Check the following to prevent inaccurate (or no) readings:

- Keep your face still and follow the on-screen guide during measurement. Do not eat, drink, or talk during measurements.
- Other apps should not be running simultaneously when using RoboVitals AI.
- Stay away from devices (e.g., microwaves, cordless phones, Wi-Fi routers) that may interfere with operation.
- RoboVitals AI must have permission to use your device camera.
- Ensure your camera is not cracked or dirty.
- Do not use other camera-interfacing apps while using RoboVitals AI.
- Ensure a stable Internet connection for RoboVitals AI to function and transmit results.
- RoboVitals AI is for individual use. Do not share the application with someone else.
- Not following on-screen instructions could result in no readings.

Note: RoboVitals AI will only report readings when there is high confidence in measurement accuracy, but this is not guaranteed. If readings cannot be generated after multiple attempts, end the session and notify your healthcare provider that the measurement was unsuccessful.

RoboVitals AI detects subtle color changes in the face as part of heart rate evaluation. Testing has shown that the report rate may be lower for darker skin tones where color changes are harder to detect.

CAUTION: Inaccurate heart rate measurements may also be caused by low arterial perfusion, excessive motion, low lighting, loss of data transmission from the camera device, or elevated heart rate greater than 103 beats per minute.

CAUTION: Inaccurate respiratory rate measurements may be caused by excessive motion, camera movement, or partial obstruction of the face, neck, or shoulders.

Specifications

MEASUREMENT	DECLARED / DISPLAYED RANGE	WINDOW	LOCATION
Heart Rate (Pulse Rate)	Displayed: 40–120 bpm · Declared accuracy range: 50–103 (±3) bpm	60 seconds	Face

MEASUREMENT	DECLARED / DISPLAYED RANGE	WINDOW	LOCATION
Respiratory Rate	Displayed: 8–30 (± 3) breaths per minute	60 seconds	Face, neck, and shoulders

Data Analysis & Limitations

RoboVitals AI measures shoulder movement as part of respiratory rate evaluation. “Belly breathing” (diaphragmatic breathing) may reduce report rate because the shoulders and thorax move less.

- Not intended as the sole method of checking physical health.
- Not for arrhythmias or irregular heartbeats.
- Not for continuous monitoring.
- Not studied in people with certain movement disorders — consult your healthcare provider if unsure.
- Unauthorized tampering with software may prevent correct operation.

Before You Start

- Use the app indoors in even lighting
- Sit comfortably with your back supported, legs uncrossed, and feet flat
- Avoid wearing face makeup or sunscreen
- Ensure your face, neck, and shoulders are clearly visible
- Relax for 5 minutes before starting
- Keep the background simple and still

During Measurement

- Position your face in the on-screen guide before and during the scan.
- Remain still and avoid talking, eating, or moving while the scan is in progress.
- Follow on-screen coaching messages and quality alerts. **Start Scan** is disabled until blocking quality issues (such as poor lighting or face position) are resolved.
- Each measurement session includes up to **three (3) scan attempts**. Attempt labels such as **2/3** appear on retries.
- Between attempts you may tap **Retry** to scan again or **Submit** to save readings you already have.

- If pulse rate is below 40 or above 120 bpm, or breathing rate is below 8 or above 30 bpm, the app will ask you to **verify** the reading by measuring again.
- When all attempts are complete, review your results and tap **Submit** to send them to your healthcare provider. Results are **not** sent until you tap Submit.
- If nothing could be measured after all attempts, only **Retry** is offered until you tap **Start Scan** to try a new session.
- After your first successful submission in a visit, **Start New Session** lets you measure again for the same patient.

After Submit

- A success message confirms your vitals were captured.
- Readings outside the normal range may show a warning banner. Tap **Retry** on the results screen if you need to verify a boundary reading.
- Tap **Done** to return home and measure again with the same Patient ID.
- If submit fails, tap **Try Submit Again**. The app automatically retries on network errors up to three times.

Troubleshooting Alerts

If RoboVitals AI does not work on your device, contact your healthcare provider for an alternative method to measure your vitals. Common on-screen alerts include:

- **Quality warnings:** resolve lighting or positioning before **Start Scan** becomes available
- **Face not detected:** center your face in the guide
- **Unable to measure:** tap **Retry** or **Start Scan** to try again
- **Submit failed:** tap **Try Submit Again**

See Section 5 for additional messages and fixes.

3. Devices and Operating Systems Compatibility

Device / Camera Requirements

- Resolution: 640×480 pixels or higher
- Frame rate: 20 fps minimum
- RAM: 6.0 GB or more

Compatible Browsers

BROWSER	VERSION
Safari	16 or higher
Google Chrome	114 or higher
Microsoft Edge	123 or higher
Samsung Internet	27 or higher

Note: Mozilla Firefox is not recommended.

Supported Operating Systems

OPERATING SYSTEM	VERSION
Android	12 or higher
iOS	18 or higher
iPadOS	17 or higher
macOS	13.7.3 (Ventura) or higher
Microsoft Windows	Windows 11 or higher

Some devices running these operating systems may not be compatible. If you have technical difficulties, contact support@hellogard.com.

| 4. Cybersecurity & Privacy

Stay Secure

- Only click links from your healthcare provider
- Never enter personal information outside the official application link
- Use private, secure Internet (avoid public Wi-Fi when possible)
- Enable security features on your device (PIN, biometrics)
- Keep your software and browser up to date

Your Data

- Your health data is encrypted during transmission and is not stored long-term on your device.
- Closing the browser ends your active session.
- For new patients, the app collects Patient ID, email, name, date of birth, height, weight, and temperature to create your record before vitals capture.
- Pulse and respiratory rate measurements are sent to your healthcare provider only after you tap **Submit**.

5. Error Messages & On-Screen Guidance

Quick reference for messages you may see during patient lookup, scanning, or submit.

Patient ID Required / Registration Incomplete

WHAT IT MEANS:

Patient ID is missing or required registration fields were not completed for a new patient.

FIX:

Enter your Patient ID and all required fields (email, name, date of birth, height, weight, temperature), then tap **Capture Vitals** again.

AI Models Still Loading

WHAT IT MEANS:

The vitals engine has not finished loading.

FIX:

Wait for the loading progress to reach 100%, then tap **Capture Vitals**. If loading fails, refresh the page or contact support.

Unable to Measure Vitals

WHAT IT MEANS:

Pulse rate, breathing rate, or both could not be measured during the session.

FIX:

Tap **Retry** (up to three attempts per session). Improve lighting, center your face in the guide, and hold still. If readings are outside the displayed range (pulse below 40 or above 120 bpm; breathing below 8 or above 30 bpm), verify by measuring again.

Verify Pulse or Breathing Rate

WHAT IT MEANS:

A reading is outside the measurable display range and needs confirmation.

FIX:

Tap **Retry** during the session, or **Retry** on the results screen after submit. You may tap **Submit** during a retry wait to save other vitals you already have.

Face Not Detected / Face Outside Guide

WHAT IT MEANS:

Your face is not centered in the guide or is not visible to the camera.

FIX:

Center your face in the frame, ensure good lighting, and tap **Start Scan** again.

Quality Warnings (Lighting, Movement)

WHAT IT MEANS:

Conditions are not ideal for an accurate scan. **Start Scan** stays disabled while a blocking warning is active.

FIX:

Adjust lighting, reduce movement, and wait for warnings to clear before starting a scan.

Camera Access Denied

WHAT IT MEANS:

RoboVitals AI cannot use your device camera.

FIX:

Allow camera access in your browser or device settings for this site, then refresh the page.

Lost Internet Connection

WHAT IT MEANS:

An Internet connection is required to load the application, authorize the session, or submit results.

FIX:

Restore your connection and try again. If submit failed, tap **Try Submit Again**.

Submit Failed / Try Submit Again

WHAT IT MEANS:

Results could not be saved to your healthcare provider.

FIX:

Tap **Try Submit Again** on the review screen. The app retries automatically on network errors. If the problem continues, contact your healthcare provider or support.

Scan Session Ended

WHAT IT MEANS:

The measurement session ended before results could be completed (for example, no remaining session attempts from the vitals engine).

FIX:

Tap **Back**, then tap **Capture Vitals** on the home screen to start a new session.

Vitals Engine Failed to Load

WHAT IT MEANS:

The browser could not load the AI measurement engine (for example, WebAssembly unavailable or memory limits).

FIX:

Use a supported browser, close other apps, refresh the page, or try another compatible device.

6. Customer Support and Contact Information

If you encounter issues or have questions, contact your healthcare provider or email Hellogard support at support@hellogard.com.

RoboVitals AI uses Informed Vital Core measurement technology. Reference eIFU: mindsetmedical.com/eifu/ivc-user-manual (<https://mindsetmedical.com/eifu/ivc-user-manual>)